

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
ORGANISATION OF ISLAMIC COOPERATION (OIC)
Department of Computer Science and Engineering (CSE)

QUIZ 1 SET B

SUMMER SEMESTER, 2023-2024

DURATION: 25 MINUTES

FULL MARKS: 20

CSE 4205: Digital Logic Design

Programmable calculators are not allowed. Do not write anything on the question paper.

Answer **all 4 (four)** questions. Figures in the right margin indicate full marks of questions with corresponding COs and POs in parentheses.

ID: _____

1. A number $(891)_b$ in an unknown base b is equivalent to the octal number $(2355)_8$. Determine the base b .

3
(CO1)
(PO1)

Solution:

Given:

$$(891)_b = (2355)_8$$

Step 1: Convert octal to decimal

$$(2355)_8 = 2 \cdot 8^3 + 3 \cdot 8^2 + 5 \cdot 8^1 + 5 \cdot 8^0 = 2 \cdot 512 + 3 \cdot 64 + 5 \cdot 8 + 5 = 1024 + 192 + 40 + 5 = 1261$$

Step 2: Convert $(891)_b$ to decimal and solve

$$8b^2 + 9b + 1 = 1261 \Rightarrow 8b^2 + 9b - 1260 = 0$$

Using the quadratic formula:

$$b = \frac{-9 \pm \sqrt{9^2 - 4 \cdot 8 \cdot (-1260)}}{2 \cdot 8} = \frac{-9 \pm \sqrt{81 + 40320}}{16} = \frac{-9 \pm \sqrt{40401}}{16} = \frac{-9 \pm 201}{16}$$

Evaluating both roots:

$$b = \frac{-9 + 201}{16} = \frac{192}{16} = \boxed{12}$$

$$b = \frac{-9 - 201}{16} = \frac{-210}{16} = -13.125 \quad (\text{Not valid})$$

Final Answer:

$$\boxed{b = 12}$$

Rubrics:

- 1 point for correctly converting the two numbers to a known base.
- 1 point for correctly expressing the given numbers into a valid equation.
- 1 point for correctly determining the root of the equation and neglecting the invalid root.

2. Convert $(43)_{10}$ into Gray Code.

3
(CO1)
(PO1)

Solution:

At first convert the given number into binary number,

$$(43)_{10} = (101011)_2$$

Final Answer:

$$(101011)_2 = (111110)_{Gray}$$

Rubrics:

- 1 point for correctly converting the decimal number into binary number.
- 1 point for writing down relevant equation/expression to convert binary number to gray code.
- 1 point for correct gray code.

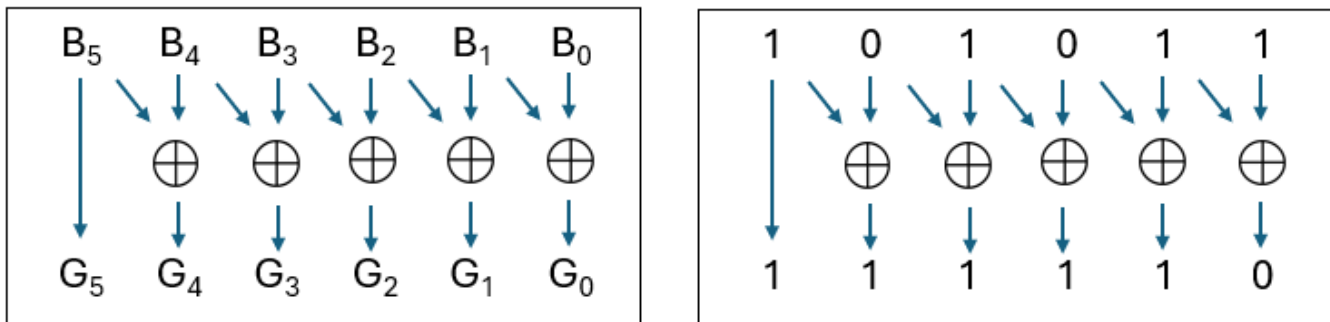


Figure 1: Binary to Gray

3. Subtract $(526)_9$ from $(857)_9$ using *Diminished Radix Complement*. (Use a 4 digit register.)

6
(CO2)
(PO1)

Solution:

Given,

$$(857)_9 - (526)_9$$

Convert to 4 digit register compatible

$$= (0857)_9 + (-0526)_9$$

Converting $(0526)_9$ to (9-1) or 8's Complement (subtract each digit from 8),

$$8\text{'s Complement of } (0526)_9 = (8362)_{8\text{'s Complement}}$$

So,

$$(857)_9 - (526)_9 = (331)_9$$

Final Answer:

$$(331)_9$$

Rubrics:

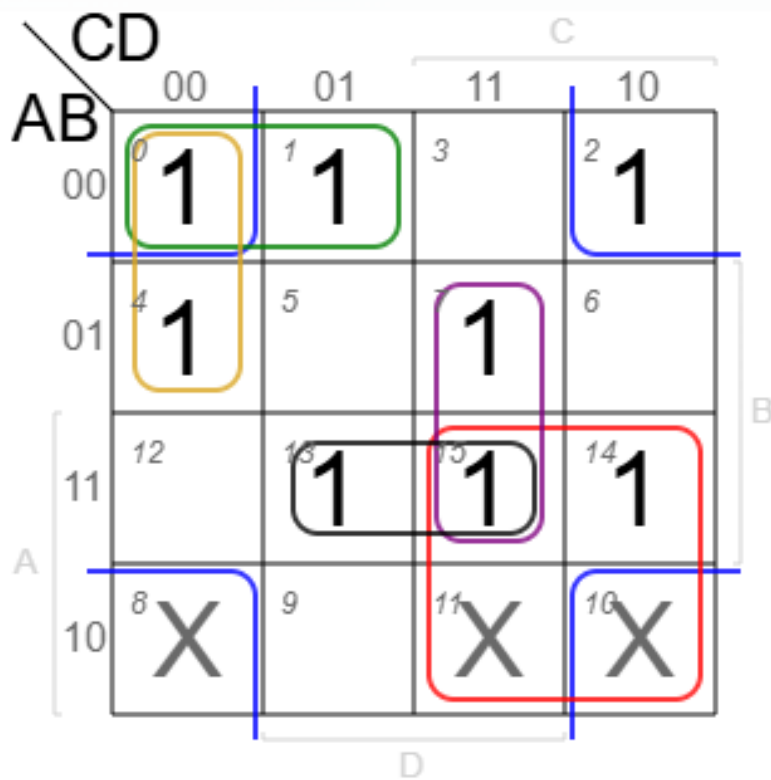
- 1 point for using 4 digit representation.
- 1 point for representing the subtraction operation in terms of addition.
- 1 point for correctly performing 8's complement.
- 2 points for correct addition using 8's complement.
- 1 point for correctly handling the overflow bit.

Solution:**Final Answer:**

$$F(A, B, C, D) = B'D' + AC + A'B'C' + A'C'D' + BCD + ABD$$

Rubrics:

- 2 point for building the 4 variable K-map.
- 1 point for correctly identifying each essential prime component (times 6).



$$F = B'D' + AC + A'B'C' + A'C'D' + BCD + ABD$$

Figure 3: K-map